

ECE 647: Lecture 4

Although the definition of a convex set only uses finite convex combinations, the property holds under more general settings.

Suppose C is convex. An infinite sequence of points $x_1, x_2, \dots \in C$. For any sequence of non-negative numbers $\theta_1, \theta_2, \dots$ such that

$$\sum_{i=1}^{\infty} \theta_i = 1.$$

We must have

$$\sum_{i=1}^{\infty} \theta_i x_i \in C.$$

whenever the summation converges.

Q) Why is this not obvious?

A)

1. We can write $\sum_{i=1}^{\infty} \theta_i x_i$ as the limit of a sequence

$$\begin{aligned} y_1 &= x_1 \\ y_2 &= \theta_1 x_1 + (1 - \theta_1) x_2 \\ y_3 &= \theta_1 x_1 + \theta_2 x_2 + (1 - \theta_1 - \theta_2) x_3 \\ &\vdots \\ y_n &= \sum_{i=1}^{n-1} \theta_i x_i + \left(1 - \sum_{i=1}^{n-1} \theta_i\right) x_n \end{aligned}$$

2. When C is convex, we know that each $y_n \in C$.
3. If C is in addition closed, then $\lim_{n \rightarrow \infty} y_n \in C$

- (a) Why? If not, then $y \triangleq \lim_{n \rightarrow \infty} y_n \in \overline{C}$ which is an open set.
 - (b) Then there must be a ball around y that is entirely in $\overline{C}$
 - (c) But that cannot happen since $y_1, \dots, y_n \in C$ and they are closer and closer to y .
4. The unusual part is, even if C is not closed, the above conclusion is still true.
 $\implies$ Convexity carries important topological properties.

Suppose a random variable X is chosen from a convex set C according to some probability distribution p .

$$\text{Then } E[X] = \int_C x p_x \in C$$

We skip the proof. Need to use the fact that any convex set is the intersection of halfspaces and hyperspaces.

1 Important examples of convex sets

In convex optimization, the constraint set must be convex.

It would be useful to identify convex sets quickly.

- 1. $\phi, \mathbb{R}^n$
- 2. Any affine set/subspace
- 3. Lines, Rays, Line Segments
- 4. Hyper-planes

$$\{x | a^T x = b\}$$

- 5. Half-spaces

$$\{x | a^T x \leq b\}$$

$$\{x | a^T x < b\}$$

- 6. Balls

$$B(x_0, r) = \left\{ x | \|x - x_0\|_2 \leq r \right\}$$

can be replaced by any "norm" that satisfies triangle inequality

7. Ellipsoid

$$E(x_0, P) = \left\{ x \mid (x - x_0)^T P^{-1} (x - x_0) \leq 1 \right\},$$

where P is a symmetric and positive semi-definite.

(Ball when $P = r^2 I$)

A is positive semi-definite if

$$x^T A x \geq 0 \forall x$$

Example: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

Any semi-definite matrix A can be written as

$$A = V \Lambda V^{-1},$$

where Λ is a diagonal matrix of non-negative diagonal elements.

8. Polyhedra

Intersection of a finite number of halfspaces and hyperplanes

$$P = \left\{ x \mid \begin{aligned} a_j^T x &\leq b_j, j = 1, 2, \dots, m \\ c_j^T x &= d_j, j = 1, 2, \dots, k \end{aligned} \right\}$$

$$P = \left\{ x \mid Ax \leq b, cx = d \right\}$$

We will see soon that convexity is preserved under intersection.

The converse is also true: Any convex set is the intersection of hyperplanes and half-spaces (could be infinitely many!)

We often describe convex sets by constraint equations:

$$\begin{aligned} h(x) &= 0, \text{ for some linear function, } h \\ f(x) &\leq 0, \text{ for some convex function, } f \end{aligned}$$

2 Operations

1. Intersection:

If C_i is a convex set for all $i \in I$, where I any index set, then $\bigcap_{i \in I} C_i$ is convex.

Useful when we laid down many constraints that need to be satisfied simultaneously.

Note: there could be unaccountably many C_i 's.

- (a) comparable to "any intersection of closed sets is a closed set."
- (b) Again, convexity seems to replace closeness.

Very useful when we keep adding constraints to an optimization problem.

- (a) Just make sure each one is convex!

2. Examples:

- (a) Polyhedra
- (b) Infinite number of intersections

The set of positive semi-definite matrices is convex.

A symmetric $n \times n$ matrix $\mathbb{X}$ is positive semi-definite if

$$V^T \mathbb{X} V \geq 0 \quad \forall V.$$

It is positive semi-definite if

$$V^T \mathbb{X} V > 0 \quad \forall V \neq 0$$

Proof: The set of positive semi-definite matrix can be written as

$$\bigcap_{V \in \mathbb{R}^n} \left\{ \mathbb{X} | \mathbb{X} \text{ symmetric and } V^T \mathbb{X} V \geq 0 \right\}$$

For each V , since $V^T \mathbb{X} V$ is a linear function of $\mathbb{X}$, hence the set

$$\left\{ \mathbb{X} | \mathbb{X} \text{ symmetric and } V^T \mathbb{X} V \geq 0 \right\}$$

is a half-space.

$$\implies \bigcap_{V \in \mathbb{R}^n} \left\{ \mathbb{X} | \mathbb{X} \text{ symmetric and } V^T \mathbb{X} V \geq 0 \right\} \text{ is convex.}$$

(c) The set

$$S = \left\{ X \in \mathbb{R}^m \mid |P_x(t)| \leq 1 \ \forall \ |t| \leq \frac{\pi}{3} \right\},$$

where $P_x(t) = \sum_{k=1}^m X_k \cos(kt)$ is convex.

Proof: For given t ,

$$\begin{aligned} & \left\{ x \mid |P_x(t)| \leq 1 \right\} \\ &= \left\{ x \mid -1 \leq \sum_{k=1}^m X_k \cos(kt) \leq 1 \right\}, \end{aligned}$$

is the intersection of two half-spaces.

$\implies S$ is convex.

Q) Is the union of convex sets a convex set?

A) Usually not.